

INTRODUCTION (L1&2)

DEFINITION:

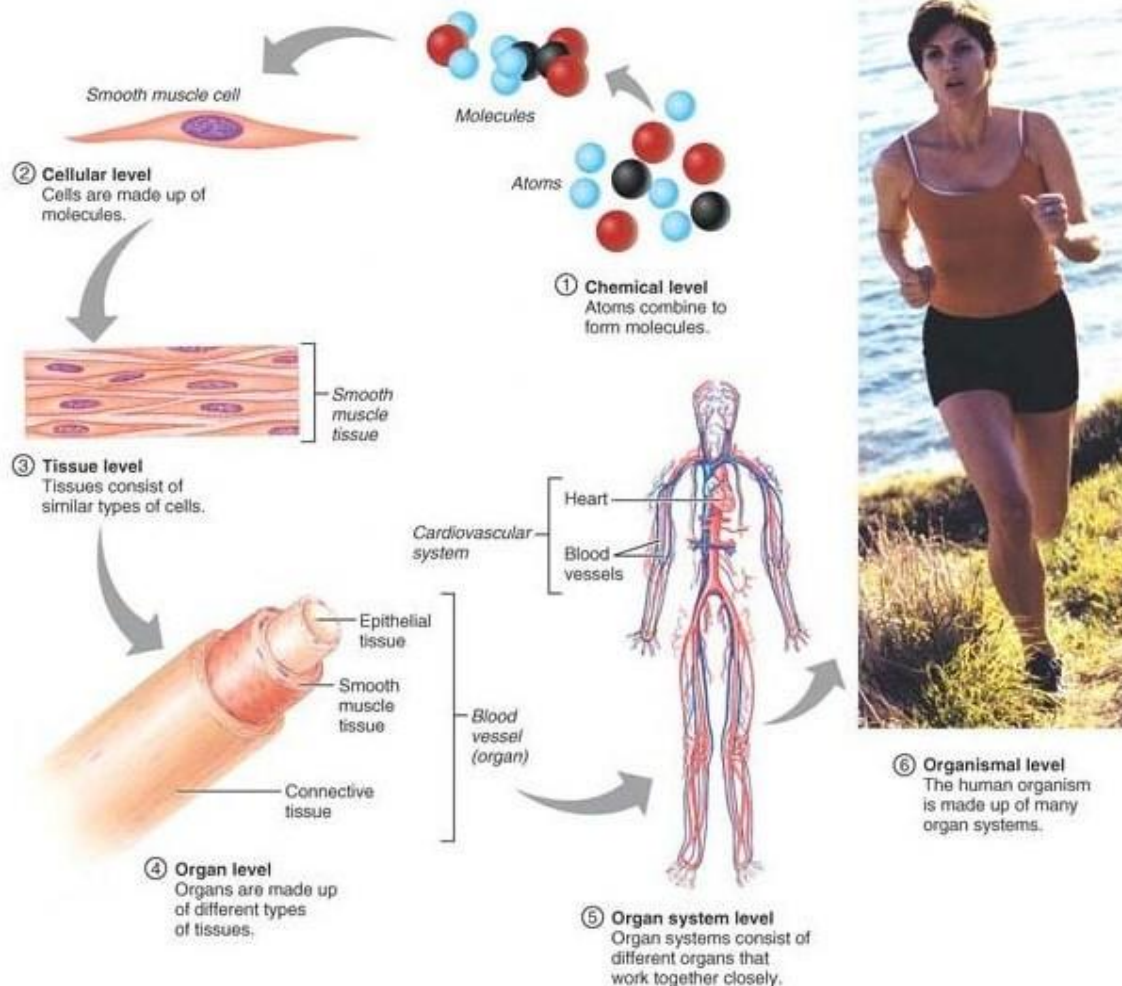
Physiology: is the study of the functioning of the normal body, and is responsible for describing *how various systems of the human body work* to maintain life and health of the individual.

Anatomy: is the study of the structure of the body and the physical relationships involved between body parts.

Levels of Organization in Body:

- The basic living unit of the body is the cell
- Each type of cell is specially adapted to perform one or a few particular functions.

The following figure showing these levels:



Composition of the human body

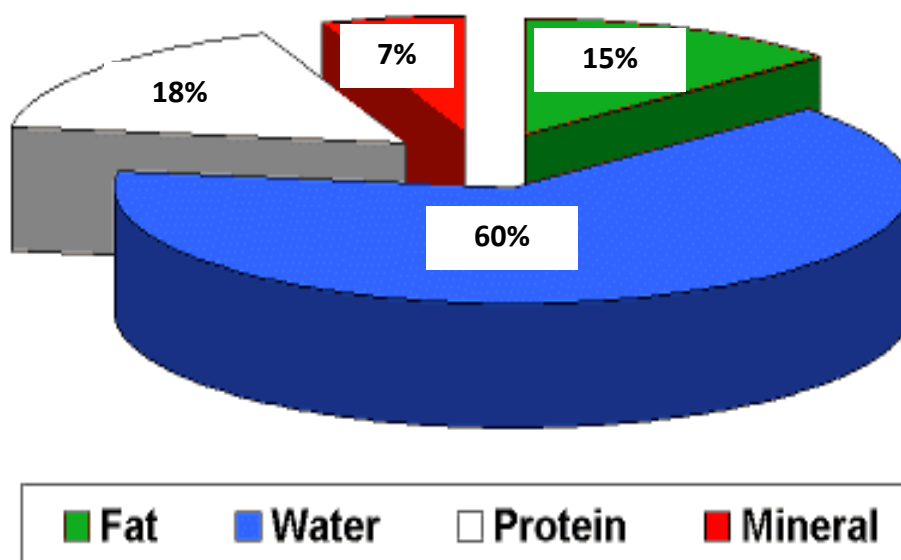
The body is composed of:

- a- **Inorganic compounds** such as water; salts; simple acids (e.g., HCl) and simple base (e.g., ammonia)

These substances are required for water balance and for many cell activities such as transporting materials through cell membranes

- b- Organic compounds** such as carbohydrates, proteins, lipids and nucleic acids. They are the structural components of the body; serve as fuel molecules that provide energy for body activities and regulate and participate in thousands of chemical reactions necessary for life

The following shape showing percentages of human body composition



TBW: means total body weight

About 98% of the body composed of only **six elements** (oxygen, carbon, hydrogen, nitrogen, calcium and phosphorus)

The most abundant elements that make up the human body and their importance showing in the following table

The following table showing elements that makes up the human body:

name	Approximate composition by mass %	Importance or functions
Oxygen	65	Required for cellular respiration; present in most organic compounds; component in water
Carbon	18	Backbone of organic molecules
Hydrogen	10	present in most organic compounds; component in water

Nitrogen	3	Component of all proteins and nucleic acids
Calcium	1.5	Structural component of bones and teeth; important in muscle contraction; conduction of nerve impulses and blood clotting
Phosphorus	1	Component of nucleic acids; Structural component of bones; important in energy transfer
Potassium	0.4	Principle positive ion within cells; important in nerve function; affects muscle contraction
Sodium	0.2	Principle positive ion in interstitial fluid; important in fluid balance; essential for transmission of nerve impulses
Magnesium	0.1	Needed in blood and other body tissues
Chlorine	0.1	Principle negative ion of interstitial fluid; important in fluid balance; component of Na Cl
Iron	Trace amount	Component of hemoglobin and myoglobin; component of certain enzymes
Iodine	Trace amount	Component of thyroid hormones

Other trace elements found in the body like Mn ; Cu; Zn; Co; F; Mo and Se

BODY FLUIDS

The fluids in the body are composed of **water** and **dissolved substances**, including electrolytes, which are crucial for body function

General functions and importance of water and electrolytes in the body:

1. temperature regulation
2. Protective cushion
3. Lubricant
4. Reactant
5. Solvent
6. Transport

Fluid compartments

There are two major fluid compartments:

- intracellular fluid (ICF) is the fluid within cells , also known as cytosol
- Extracellular fluid (ECF) is the fluid found outside of cells

There are two major kinds of extracellular fluid

- interstitial fluid is the fluid surrounding the cells
- Plasma is the fluid component of blood

Body Water

* Total Body water (TBW) = **60 %** of the total body weight.

- **Intracellular fluid (ICF 40 %).**

- **Extracellular fluid (ECF 20%) which composed of:**

1. Interstitial fluid (ISF 15 %).
2. Plasma (5 %).

Total body water distribution:

1. Muscle 50 %
2. Skin 20 %
3. Other organ 20 %
4. Blood 10 %

Water Balance

The following table showing the source of both input and output of water

Daily Water "Balance"					
Input			Output		
Metabolism		250 mL	Sweat		200 mL
Moist foods		750 mL	Feces		100 mL
Drinking		1500 mL	Skin& Lungs		700 mL
-		-	Urine		1500 mL
TOTAL		2500 mL	TOTAL		2500 mL

Physiological Variations in TBW :

- 1 - It is greater in young infants (75% of his total body weight).
- 2 - It is less in females because of the higher fat content.
- 3 - It is less in obese peoples.
- 4 - It is less in old age.

The following table showing the percentage of body water in different ages.

Total Body Water	
Newborn infant	75-80% of total body weight
Adult male	60%
Adult female	45-50%
Elderly	50%

Effects of water loss:

- Loss of 1 % of TBW ---> Thirst sensation.
- Loss of 5 % of TBW ----> Dehydration.
- Loss of more than 20 % of TBW ----> Death.

Electrolytes:

An ionic compound, e.g. sodium chloride, in solution water is called an electrolyte because it can conduct electricity.

Electrolytes are important body constituents because:

- some conduct electricity, essential for muscle and nerve function
- some exert osmotic pressure, keeping body fluids in their own compartments
- some function in acid-base balance, as buffers to resist pH changes in body fluids.

A large number of compounds present in the body are not ionic and therefore have no electrical properties when dissolved in water, e.g. carbohydrates.

Distribution of electrolytes in the body:

The major **cations** of ICF are K^+ and Mg^{2+} while the major **anions** of ICF are protein and organic phosphates [adenosine triphosphate (ATP), adenosine diphosphate (ADP), and adenosine monophosphate (AMP)].

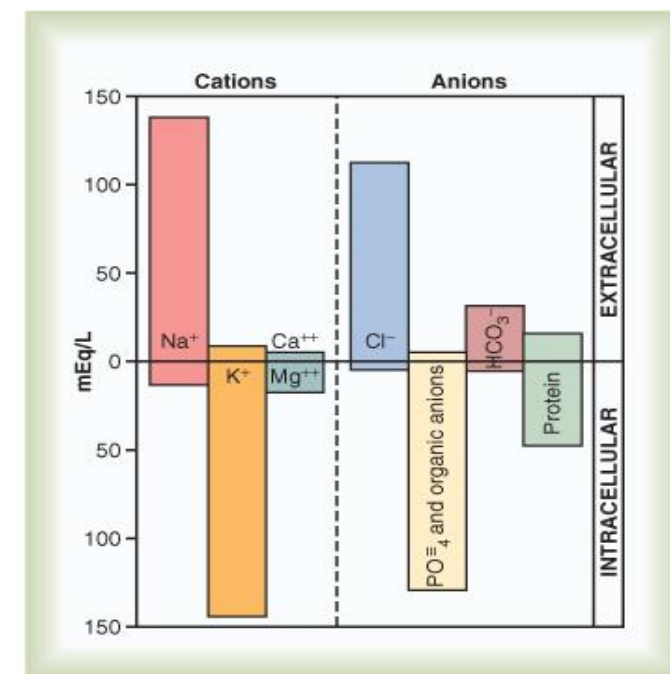
The major **cation** of ECF is Na^+ while the major **anions** of ECF are Cl^- and HCO_3^- .

- The composition of the ECF is maintained by the cardiovascular, pulmonary, renal, gastrointestinal, endocrine and nervous system acting in a coordinated fashion.

- The composition of the ICF is maintained by **cell membrane**, which mediates the transport of materials between the ICF and ECF by diffusion, osmosis, active transport, and vesicular transport

The following shapes showing the amount of cations and anions in both of ICF and ECF.

	EXTRACELLULAR FLUID	INTRACELLULAR FLUID
Na ⁺	142 mEq/L	10 mEq/L
K ⁺	4 mEq/L	140 mEq/L
Ca ⁺⁺	2.4 mEq/L	0.0001 mEq/L
Mg ⁺⁺	1.2 mEq/L	58 mEq/L
Cl ⁻	103 mEq/L	4 mEq/L
HCO ₃ ⁻	28 mEq/L	10 mEq/L
Phosphates	4 mEq/L	75 mEq/L
SO ₄ ⁻	1 mEq/L	2 mEq/L
Glucose	90 mg/dl	0 to 20 mg/dl
Amino acids	30 mg/dl	200 mg/dl ?
Cholesterol	0.5 g/dl	2 to 95 g/dl
Phospholipids		
Neutral fat		
PO ₂	35 mm Hg	20 mm Hg ?
PCO ₂	46 mm Hg	50 mm Hg ?
pH	7.4	7.0
Proteins	2 g/dl (5 mEq/L)	16 g/dl (40 mEq/L)



Homeostasis

Definition: is keeping the composition of the internal environment constant (ECF).

- It is essential for normal cell function.

The ECF is the same in different parts of the body because:

- 1- Rapid exchange of solutes and water between blood and interstitial fluid.
- 2- Rapid circulation of blood.
- 3- Very near capillary to any cell.

Factors homeostatically regulated include:

1. Concentration of nutrient molecules.
2. Concentration of O₂ and CO₂.
3. Concentration of waste products.
4. pH
5. Concentration of water, salt, and electrolytes.
6. Volume and pressure.
7. Temperature.

Homeostatic regulation involves three parts:

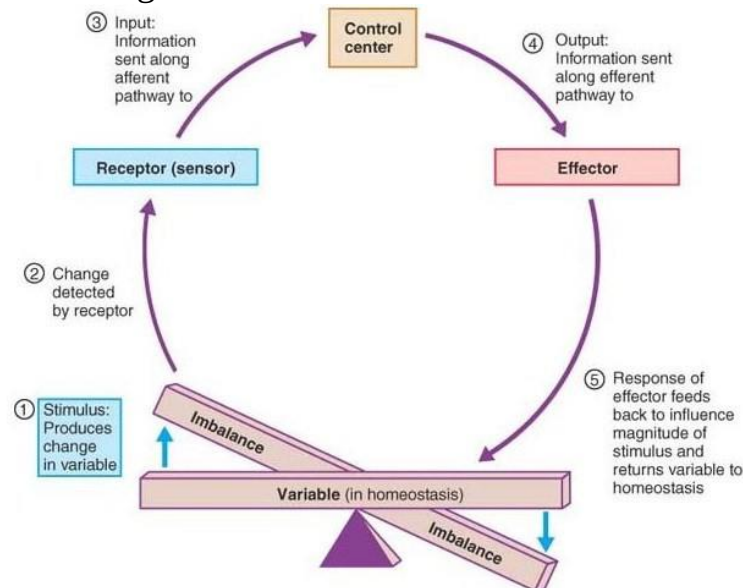
- 1) The receptor
- 2) The control center
- 3) The effector.

The receptor receives information that something in the environment is changing.

The control center or integration center receives and processes information from the receptor.

The effector responds to the commands of the control center by either opposing or enhancing the stimulus.

The following figure showing the elements of a homeostatic control system



- Communication between the receptor, control center, and effector is essential for normal operation of the control system.

Homeostatic Control Systems:

Control systems are grouped into Two Classes:

1. Intrinsic controls

- Local controls those are inherent in an organ.
- Operate in all cells to help control intracellular function as well as extracellular function

2. Extrinsic controls

- Regulatory mechanisms initiated outside an organ.
- Operate throughout the entire body to control the interrelations between the organs.
- Accomplished by nervous and endocrine systems.

Control of homeostasis:

1- Nervous mechanisms:

It is a rapid regulatory mechanism that regulates body functions by nerve impulses carried by nerve fibers.

e.g.: ABP (arterial blood pressure) regulation by vasomotor center (VMC).

2- Hormonal mechanisms:

It is a slow regulatory mechanism that regulates body functions by hormones released from endocrine glands. e.g.: To maintain glucose level constant.

Feed back control of homeostatic mechanisms:

A)- Negative feedback control:

- In which the resultant effect of a function, inhibit the resultant function.

(opposes an initial change)

- Most control systems of the body act by negative feedback

e.g.: 1- Regulation of Co₂ in blood by lung

When Co₂ increases in arterial blood, excites the respiratory center, causing the pulmonary ventilation increases. This increases expiration of carbon dioxide and, therefore, removes excess carbon dioxide from the blood and tissue fluids. This process continues until the concentration returns to normal.

2- ABP regulation by (VMC):

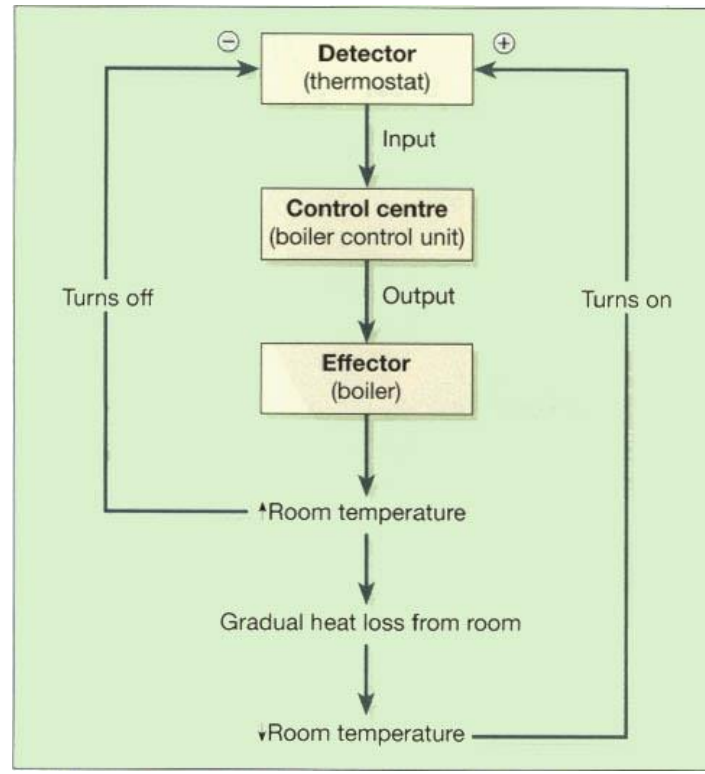
Increase ABP above normal, inhibits the VMC which inhibit the heart pumping activity and causes vasodilatation of the blood vessels. Both of these effects decrease the arterial pressure back toward normal.

3- Regulation of blood glucose level:

If blood glucose is increased above normal, insulin hormone is secreted to maintain its normal level.

4- Control of body temperature (like control of room temperature by a domestic boiler)

The following shape showing mechanism of negative feedback control



Our bodies control body temperature in a similar way.

The brain is the **control center**, **the receptor** is our body's temperature sensors, and **the effector** is our blood vessels and sweat glands in our skin.

When we feel heat, the temperature sensors in our skin send the message to our brain.

Our brain then sends the message to the sweat glands to increase sweating and increase blood flow to our skin.

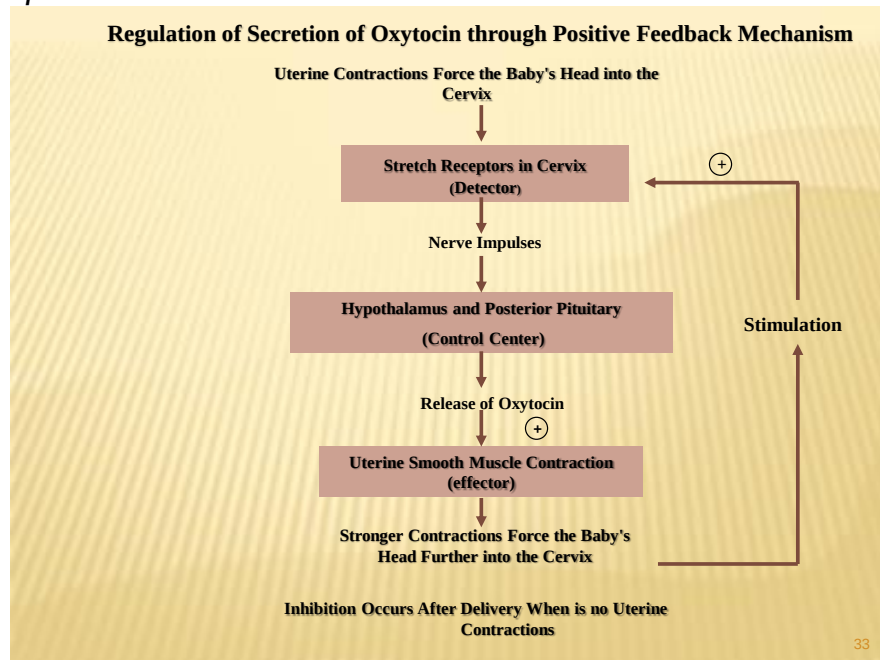
When we feel cold, the opposite happens. Our brain sends a message to our sweat glands to decrease sweating, decrease blood flow, and begin shivering.

B- Positive feedback control:

In which the resultant effect of a function, stimulates the resultant function.

(amplifies an initial change)

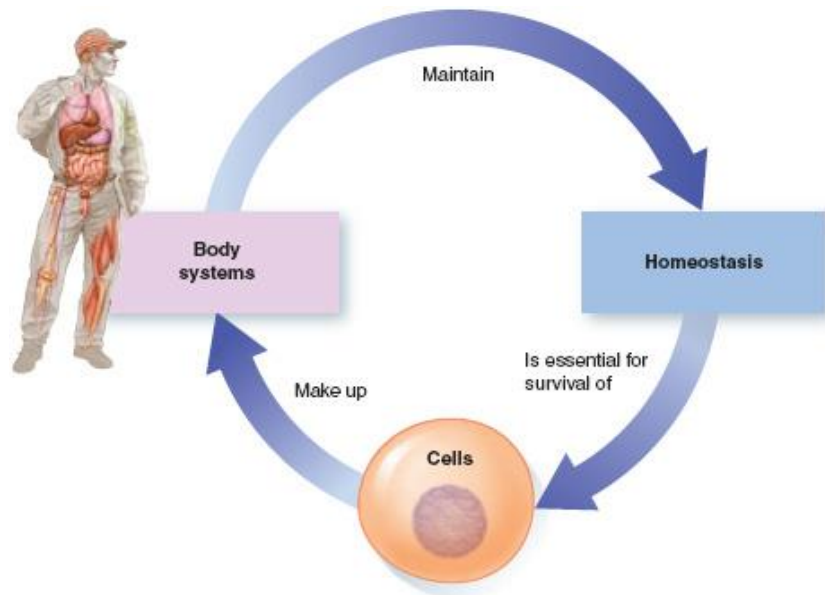
e.g.: **1- Parturition:** in which the descent of the baby, stretches the uterine cervix that intern causes reflex uterine contraction and more descent and so on.



The organ systems work together to maintain life

Table 1.4 Organ Systems of the Body		
System	Major Organs	Primary Functions
Integumentary	Skin, hair, nails	Protection, thermoregulation
Nervous	Brain, spinal cord, nerves	Regulation of other body systems
Endocrine	Hormone-secreting glands, such as the pituitary, thyroid, and adrenals	Secretion of regulatory molecules called hormones
Skeletal	Bones, cartilages	Movement and support
Muscular	Skeletal muscles	Movements of the skeleton
Circulatory	Heart, blood vessels, lymphatic vessels	Movement of blood and lymph
Immune	Bone marrow, lymphoid organs	Defense of the body against invading pathogens
Respiratory	Lungs, airways	Gas exchange
Urinary	Kidneys, ureters, urethra	Regulation of blood volume and composition
Digestive	Mouth, stomach, intestine, liver, gallbladder, pancreas	Breakdown of food into molecules that enter the body
Reproductive	Gonads, external genitalia, associated glands and ducts	Continuation of the human species

The following figure showing Interdependent relationship of cells, body systems, and homeostasis



- Homeostasis is essential for the survival of cells.
- Body systems maintain homeostasis, and cells make up body systems.
- This relationship serves as the foundation for today physiology

Disruptions in Homeostasis:

- ❖ ***Can lead to illness and death.***
- ❖ ***Pathophysiology***
Abnormal functioning of the body associated with disease.